Topic 2.1 Basic Concepts of ICT

OBJECTIVES

At the end of this c, you should be able to:
1. Be familiar with what is a CPU and memory;
2. Explain data representation and coding scheme;
3. Be familiar with various tools and hardware inside the system unit;
4. Be familiar with various input methods and devices;
5. Be familiar with various output methods and devices;
6. Be familiar with and explain three types of magnetic storage;
7. Be familiar with and explain four types of optical storage; and
8. Be familiar with the advantages of secondary storage.

2.1 CENTRAL PROCESSING UNIT (CPU)

Introduction

In this topic, we will discuss the Central Processing Unit (CPU). The CPU or the processor is the center of all activities responsible for changing raw data into the required information inside the computer.

The CPU is a collection of complex electrical circuits containing thousands or millions of transistors, which are placed or installed over an integrated circuit. The integrated circuit is also called the chip or microchip, because all the transistors are designed or placed on a very small silicon chip.

2.1.1 Central Processing Unit (CPU)

As we know, CPU is the heart of a computer. Do you know what components are inside the CPU?

The CPU or processor is designed and built in various forms. In the personal computer, a processor is a microprocessor chip. In a bigger system, such as the mainframes and the supercomputers, a few circuit boards are used. The CPU can be present in various forms, and is placed on the motherboard or system board.

The job of a CPU is to execute the stored software instructions, turning data into information as shown in Figure 2.1. These instructions are executed in machine language, as shown in Figure 2.1 below. CPU is divided into two main sections; these are
(i) Control Unit, and
(ii) Arithmetic Logic Unit (ALU).

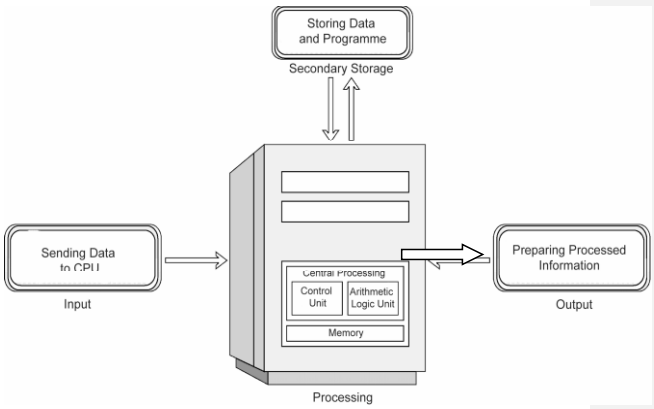


Figure 2.1: Control

Control Unit
The Control Unit of the processor can be compared to the "human brain". It informs the entire computer on how to execute programme instructions. The job of the control unit is shown in Figure 2.2 below:

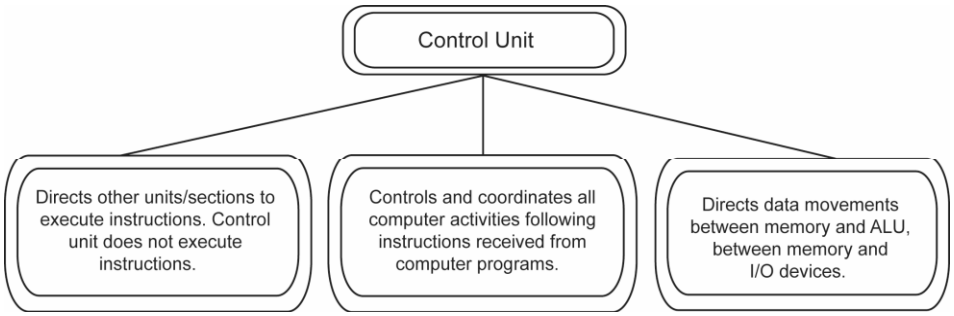
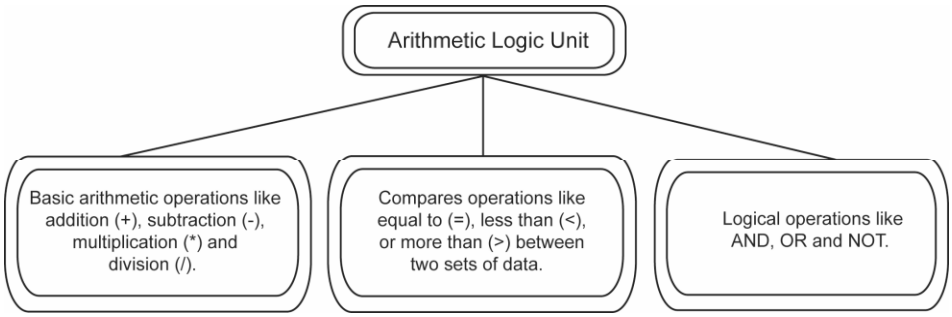


Figure 2.2: Control Unit

In other words, the control unit manages the taking of an instruction, decodes the instruction to know its aim, instructs the ALU to execute the instruction, and then directs storing of results to the memory (primary storage), register or cache.

Arithmetic Logic Unit
Arithmetic Logic Unit (ALU) executes two types of operations; which are arithmetic operation and logical operation, as shown in Figure 2.3 below:



2.1.2 Microprocessor Chip
Inside bigger computers, such as the mainframes and supercomputers, various CPU functions are carried out by chips on different circuit boards. However, for microcomputers, the CPU is usually located only on one chip, which is termed as the microprocessor. Besides containing the control unit and ALU, the microprocessor also contains registers and a system clock.

Most new microprocessors have been integrated with multimedia technology or 3D graphics. Specific microprocessors for notebooks and servers are also being made and sold by manufacturers.

Two types of Microprocessor Chips

(i) **CISC Chip:** Microprocessors that are commonly used are the Complex Instruction Set Computing (CISC) chip. CISC is a type of computer architecture with a large number of instructions that require hard coding approach. This technology has been popularised by Intel, and has become the basis of its microprocessors. This chip design has been used very widely, and many programmes have been written specifically for this type.

(ii) **RISC Chip**: This type of computer chip uses the approach of Reduced Instruction Set Computing (RISC), which involves less instruction. Its design is simpler and cheaper than the CISC chip. RISC is a type of architecture in computing that operates with a relatively small set of instructions for its coding approach. This chip is used mostly inside high-powered microcomputers. Refer to Table 2.1 for the most popular microprocessors.

Table 2.1: Popular Microprocessors

Microprocessor	**Type**	**Normal Uses**
Pentium	CISC	Microcomputers
PowerPC	RISC	Apple Macintosh computer
Alpha	RISC	Supercomputers, Workstations
MIPS	RISC	Workstations, Video Players

Commented [o1]: Reduced instruction set computing; reduced instruction set computer; RISC: (computer science) a kind of computer architecture that has a relatively small set of computer instructions that it can perform
/
This RISC is a type of architecture in computing that operates with a relatively small set of instructions for its coding approach

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complex instruction set computing; complex instruction set computer; CISC: (computer science) a kind of computer architecture that has a large number of instructions hard coded into the CPU chip
/

This CISC is a type of computer architecture with a large number of instructions that require hard coding approach

2.1.3 Memory

The memory is separated from the CPU. But the memory is normally integrated with the CPU and accessed by it. Data and instructions need to be present inside the memory before being accessed by the CPU. The memory simply holds the data and instructions temporarily, as long as the programme being used is operational. In general, the storage space for the memory can be used in five different forms or functions and these are the following:

1) For storing the operating system and other system software programmes that act as interfaces between hardware and application programmes and controlling computer resources.
2) For storing application programmes like word processors, spreadsheets, and databases.
3) For storing data temporarily, i.e. data received from input devices or secondary storage to be processed or to be sent to output devices or secondary storage after processing.
4) For storing data required during processing inside an area of storage work.
5) For providing additional storage space to programmes or data, if required.

If the computer has excessive storage space for memory, the excess space is not to be used, but it can or is ready to be used. The total memory utilisation changes during processing.

Commented [o2]: The memory utilisation changes during processing. This is because
a)The system assigns perment storage space for each programme when it is activated
b) All programmes use the same size of storage space when activated
c)Each programme storage space for execution is assigned atomatically
d)Users normally adjust storage space to suit their running programme

The memory communicates with the processor through:

1) Address Bus - driven by the processor;
2) Data Bus - driven by the processor for the write cycle, but driven by the memory for the read cycle;
3) Signal/Control Bus - for determining whether to read/write, when to start, data validity, and others.

Commented [o3]: The type of bus in the system unit that supports the processor for the write cycle, but driven by the memory for the read cycle is called…
-Data bus

The main memory communicates with the procerssor via bus where the actions are controlled by the processor is called…
-address bus

The bus type for communication between the main memory and the processor that determes whether to read or write, when to start, and data validity is called
-Control or signal bus

There are three types of memory chips as shown in Figure 2.4 below, which are, Random Access Memory (RAM), Read Only Memory (ROM), and Chip Metal Oxide Semiconductor (CMOS).

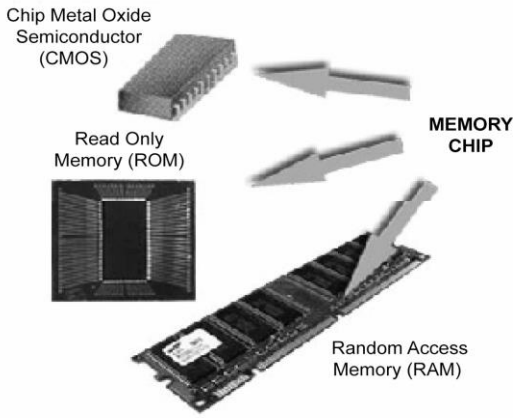


Figure 2.4: Three types of memory chips

RAM

The Random Access Memory (RAM) stores the programmes and data being processed by the CPU. It is a temporary storage because as soon as the electrical switch is turned off, all that has been saved inside the RAM will be lost. However, there is a new type of RAM that can store programmes and data permanently. The Flash RAM, or Flash memory chip can store data even when there is an electrical failure. This type of memory is more expensive than the normal RAM, and is mostly used in portable computers. The capability or content of the RAM is measured in bytes. Four units of measurement that are normally used to describe memory capability are as shown in Table 2.2 below:

Table 2.2: Memory Capacity

Unit	**Capacity**
Kilobyte (KB)	1,024 bytes
Megabyte (MB)	1,024,000 bytes
Gigabyte (GB	1,024,000,000 bytes
Terabyte (TB	1,024,000,000,000 bytes

Two types of RAM chips that are normally used are:

(i) **DRAM,** which stands for Dynamic RAM, is the memory that holds data in a dynamic manner. Data is not stored continuously. Instead, the computer will refresh the data hundreds of times every second. DRAM is normally used in personal computers.

(ii) **SRAM,** which stands for Static RAM, is the memory that holds data continuously for as long as there is electricity, without any refresh process. SRAM is faster than DRAM, but its design is more complex.

The ROM chip contains a programme that is loaded in the factory. The ROM chip does not evaporate and cannot be changed by users. “Read-Only” means that the CPU can read or access the programme written inside the ROM chip. Computers cannot write, code or change any data or instruction inside the ROM. There are four types of ROM that are normally used, as shown in Table 2.3 below.

Type of ROM	**Functions**
PROM	Programmable ROM
EPROM	Erasable Programmable ROM, the content of which can be erased by using the Ultra-Violet Ray method
EEPROM	Electrically EPROM - PROM that can be erased by electricity, the content of which can be changed or programmed electronically
Flash Memory	Also known as Flash ROM - Flash Memory can be erased and programmed electronically.

CMOS

The CMOS chip provides flexibility and expandability to the computer system. It contains important information that is required every time the computer system is started. This chip provides information such as the total RAM size, types of keyboard, mouse, monitor, and disc drive. It uses battery power and its content remains permanent even after the electricity supply is disconnected. Its content changes according to changes inside the computer system, such as the addition of new hardware and RAM.

Disk/Storage Drive

Drive is a media where information can be processed. The standard drives are C:, D:, and E:

How to Format or Initialize a Disk/Drive

Format is a command used to prepare a drive or disk to accept information or to erase entire information on a storage device. Formatting must be used with caution, because it erases everything on the storage media.

Unit of measurement of storage media

- 1,024 Byte = 1 kilobyte (KB)
- 1,024,000 Byte = 1 megabyte (MB)
- 1,024,000,000 Byte = 1 gigabyte (GB)
- 1,024,000,000 ,000 Byte = 1 terabyte (TB)

Computer Definition

What is a computer? It is an electronic device which performs high-speed arithmetic and makes a logical decision under the control of instructions supplied by human. In addition, it is automatic and store information for future use.

The Arithmetic or Machine Language

The only way in which data can be represented in the computer is in the form of bits (**BI**nary digi**TS**). That is all data in computing or information systems is stored as bits on a storage device. Eight (8) bit represents one byte or character. The byte is used to measure the capacity of the storage device space.

Symbolic language

G	O	A	L

Machine Language

0	0	0	0
1	1	1	1
0	0	0	0
0	0	0	0
0	1	0	1
1	1	0	1
1	1	0	0
1	1	1	0

2.1.4 Data Representation

We know that people interact by using language. How do computers interact?

Computers consider all data, numbers, alphabets, and symbols as numbers. Even computer instructions are also in numerical form (machine language). The numbering systems used by computers are the binary numbers, hexadecimal numbers, and octal numbers, as shown in Table 2.4 below. Computers use binary digits - zero (0) and one (1) - normally called bits, to represent data.

Table 2.4: Numbering System Symbols

Numbering System	**No. of Symbol**	**Symbols Used**
Binary System	2	0, 1
Octal System	4	0, 1, 2, 3, 4, 5, 6, 7
Decimal System	8	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Hexadecimal System	16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

There are three binary coding schemes - the most popular one uses eight (8) bits to form one (1) byte. These codes are ASCII, EBCDIC, and Unicode, as explained in Figure 2.5 below.

Code	Definition
ASCII	American Standard Code for Information Interchange.
EBCDIC	Extended Binary Coded Decimal Interchange Code
Unicode	16-bit code invented to cater for the need of international languages such Chinese and Japanese

When a key is pressed on the keyboard, the character is changed automatically to a series of electronic pulses that can be recognised by the system. For example, when the key A is pressed, it will cause an electronic signal to be transmitted to the microcomputer system unit. The system unit will then change the signal into ASCII code, which is, 0100 0001. Before execution, all instructions and data need to be converted into binary form. For example, 3 + 5 requires 24 bits with the use of ASCII coding scheme. It would be good if we can understand the concept of converting numbers between binary, octal, and hexadecimal systems as illustrated in Table 2.6 below.

Table 2.6: Examples of Numbers in Various Numbering Systems

Decimal	Binary	Octal	Hexadecimal
1	0001	01	01
2	0010	01	01
5	0101	05	05
8	1000	10	08
10	1010	12	0A
11	1011	13	0B
15	1111	17	0F
16	0001000	20	10

Please refer to Table 2.7 below to see how data is represented in ASCII and EBCDIC codes.

Table 2.7: Binary Coding Schemas for ASCII and EBCDIC

Symbol	ASCII	EBCDIC
A	0100 0001	1100 0001
B	0100 0010	1100
C	0100 0011	1100
D	0100 0100	1100
E	0100 0101	1100
F	0100 0110	1100
G	0100 0111	1100
H	0100 1000	1100
I	0100 1001	1100 1001
J	0100 1010	1101
K	0100 1011	1101
L	0100 1100	1101
M	0100 1101	1101
N	0100 1110	1101
O	0100 1111	1101
P	0101 0000	1101
Q	0101 0001	1101
R	0101 0010	1101
S	0101 0011	1110
T	0101 0100	1110
U	0101 0101	1110
V	0101 0110	1110
W	0101 1011	1110
X	0101 1000	1110 0111
Y	0101 1001	1110
z	0101 1010	1110

Symbol	ASCII	EBCDIC
!	0100 0001	1100 0001
“	0100 0010	1100
#	0100 0011	1100
$	0100 0100	1100
%	0100 0101	1100
(	0100 0110	1100
)	0100 0111	1100
*	0100 1000	1100
+	0100 1001	1100 1001
0	0100 1010	1101
1	0100 1011	1101
2	0100 1100	1101
3	0100 1101	1101
4	0100 1110	1101
5	0100 1111	1101
6	0101 0000	1101
7	0101 0001	1101
8	0101 0010	1101
9	0101 0011	1110

2.1.5 System Unit

The System Unit is a hardware component or a space where the processor, memory chips, ports, buses, additional slots, board (hardware in which chips and related circuits are placed), and other electrical components are located. It is also called the “System Cabinet” for the mainframes, or the “System Board” for the microcomputers. Even though the system unit also includes hard disks, floppy disks, CD-ROM, and others, we are not going to touch on these devices, as they are located outside the CPU. We are only going to focus on the sections listed below:

Power Supply
The power supply plays the role of changing electrical current from Alternating Current (AC) to Direct Current (DC). Normally, there is a fan to cool down the transformer and other components. If the electrical power is not uniform, e.g. there is high-and-low voltage or power surge, this may spoil the main board, hard disks, and others. Hence, it would be better to install special equipment such as Uninterrupted Power Supply (UPS), voltage regulator, and so on.

Main/System Board
The system board, or main board, is a communication network for the entire computer system. Every component of the system unit is connected directly to the system board. It functions as a data-bus that enables various components to communicate with each other. External devices such as the keyboard, mouse, and monitor cannot communicate with the system unit without the system board.

The system board is a thin circuit board that is filled up with sockets and electronic components including various types of chips. One chip contains a very small circuit board embedded in a stamp-size piece of silicon. This chip is also known as the silicon chip, semiconductor or integrated circuit. The chips are packaged and inserted into the sockets of the system board. These electronic components and chips are fixed to the system board as illustrated in Figure 2.5.

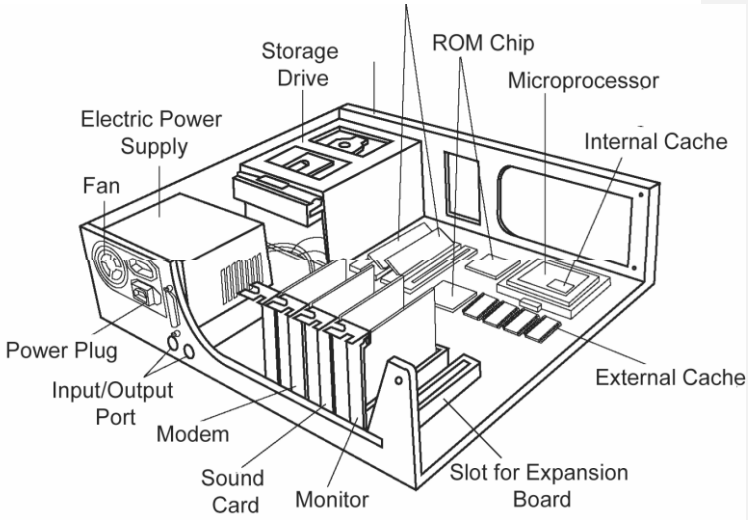


Figure 2.5: Main board

ROM Chip
The ROM chip contains a programme that has been developed or burnt at the factory. Thus, the programme that is required for computers to operate, or to start operating the computer - such as hardware checking, BIOS, and so on. BIOS stands for "Basic Input Output System". The BIOS chip is normally used, and is partly hardware and software.

The BIOS provides a service enabling software to communicate with the input and output devices. The ROM BIOS contains specific instructions. When a computer is switched on, the BIOS will perform "Power-On Self-Test" (POST), such as diagnostic tests for the CPU and memory. It will then test communications with the hardware such as the keyboard, disk drive, and others.
Finally, the BIOS will boot up the operating system and submit control to the operating system.

Slot and Expansion Board

Many computers are of the open architecture, where we can open up and add-on new devices, and then expand its capability. Expansion means adding more memory or devices or software. This is made possible through the expansion slot and the expansion board. The expansion slot is a socket on the main board, which enables an expansion card to be installed. An expansion card, expansion board, adapter or simply card, is actually a circuit board that gives more memory, or input/output device control or software.

An expansion card is inserted into the slot inside the system unit. A port on the card enables cabling to be connected from the added card to the device outside the system unit. We will see various functions of the boards or cards, as shown in Table 2.8.

Table 2.8: Functions of Boards or Cards

Board or Card	**Functions**
Network Adapter Card	Connects one computer to other computers. This is a communication network in which users can share data, programmes and hardware. This card normally connects the system unit to a cable that connects other devices in the network. The network adapter card is inserted into the slot inside the system unit.
Small Computer System Interface (SCSI) Card	Most computers have limited expansion slots. The SCSI card uses only one slot, but it can connect up to seven devices to the system unit. This card is used to connect devices like the printer, hard disc drive and CD-ROM to the system unit
TV Adjuster Card	The TV board that contains the TV and video adjuster can change the TV signals to a form that can be displayed on the monitor.
PC Card	Due to the size and limitations of portable computers, an additional board of credit-card size has been created. This card can easily be slotted in and changed from the outside of portable computers. This is called the PC Card, or the Personal Computer Memory Card International Association (PCMCIA).
Memory Expansion	It is useful for increasing memory and associating with other computers. Enables addition of RAM chips.
Display Adapter Card	Enables various colour video display for monitors being used.
Control Card	Enables the CPU to work with various software devices such as the disk control card.
Accelerator Card	Increases the speed of computers
Emulator Card	Enables computers to emulate other devices like the terminals.
Fax-Modem card	Receives fax/data/information via telephone lines
Graphic Adapter	Enables computers to use a certain graphic standard.
Sound Card	Translates data into sound output.
Voice Card	Enables voice output.

Port

The port is a connecting socket located outside the system unit. The port enables software devices, or input/output devices to be plugged in for connection to the computer, so that they can communicate with the computer system. A cable can be used to connect input/output devices to the system unit through the port, as shown in Figure 2.6

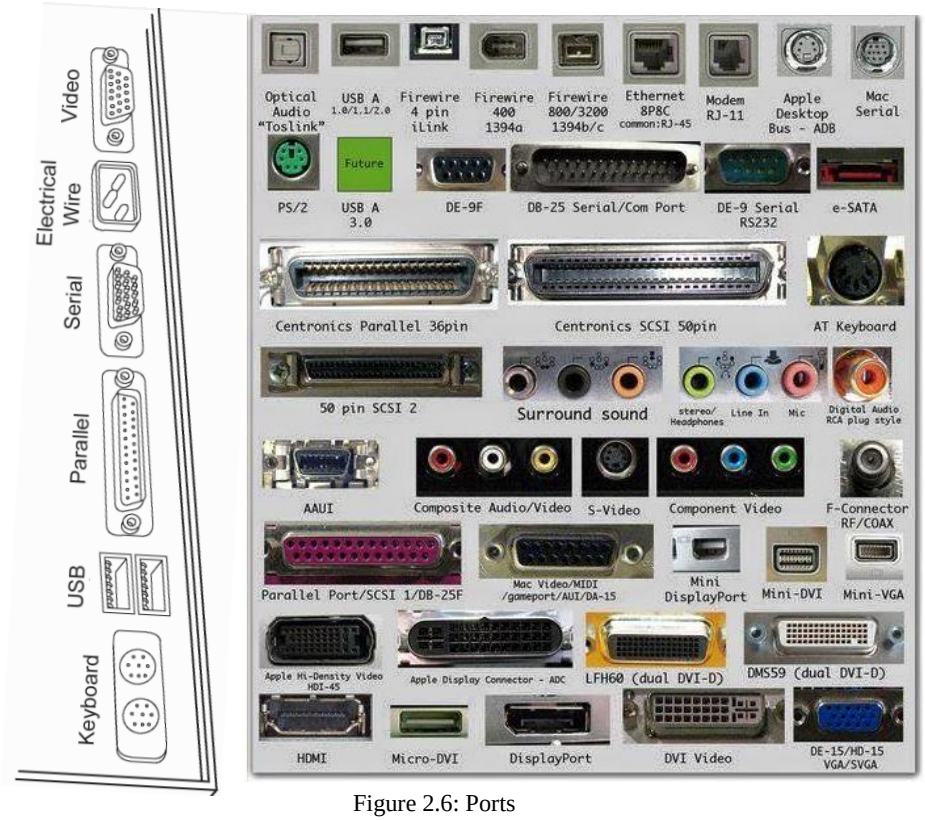


Figure 2.6: Ports

There are various types of ports and some of their functions are as shown in Table 2.9 below.

Commented [o4]: The type of computer port that supports users to plug in (attach) an external device without rebooting the system unit is called
A)USB
B)Serial
C)Parallel
D)Keyboard

Table 2.9: Various Types of Ports

Types of Ports	Functions
Parallel Port	It is used for connecting external devices, which sends or receives data in a big amount at a short distance. This port normally sends simultaneously 8-bit data via 8 parallel wires. A parallel port is used to connect a printer to the system unit.
Serial Port	It connects the mouse, keyboard, modem, and other devices to the system unit. Serial port sends data 1 bit at a time, and is suitable for sending information at long distance.
Accelerated Graphic Port (AGP)	It is used for connecting to the monitor, and can support high-speed graphics and other video inputs.
Universal Serial Bus (USB) Port	It is forecasted to replace serial and parallel ports. It is fast, and one USB port can connect a few devices to the system unit. The latest USB (Type C) devices/ports run up to 40 Gbps.
Electrical Wire Port	The latest port that is faster than the USB port, and is used to connect high-speed printers and video cameras to the system unit.

High Definition Media Interface (HDMI)	HDMI is one of the latest digital interface port that connects high definition and ultra-high definition devices such as projectors, monitors, HDTVs, cameras, and gaming consoles. Also, it can be used to carry uncompressed video and audio signals. The latest HDMI (2.1) devices operate at 48 Gbps.

2.1.6 System Clock

The System clock is a circuit chip, which generates electronic pulses at a fixed rate to synchronise or control timing activities of processing. The control unit is very dependent on the system clock for doing its jobs. The system clock controls the speed of operation inside the computer. This speed is measured in megahertz (MHz). One megahertz is equal to one million cycles per second. The faster the clock speeds, the faster the computer's ability to process information.

2.1.7 Bus Line

Try to think how data moves inside the computer.

The bus line is also known as data bus, or simply "bus", which connects the CPU components to each other. It also connects the CPU to various other components on the system board. The bus is a data passage way along which bits move. This data passage way is similar to the highway. The more lanes it has, the faster the traffic moves. Similarly, the bigger the bus capacity, the faster the speed of the computer execution. Figure 2.7 shows the bus line on the CPU.

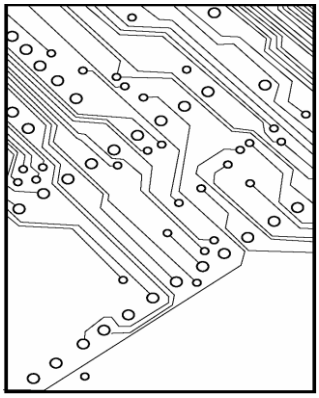


Figure 2.7: Bus Line

When a microprocessor chip changes, the bus line also changes. Most of the devices like the expansion board, will work with one type of bus only. There are three types of bus lines as shown in Figure 2.8

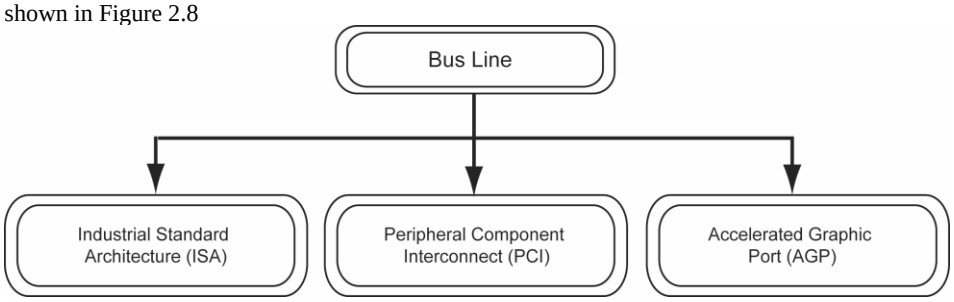


Figure 2.8: Three bus lines

i. Industrial Standard Architecture (ISA) bus was built for the IBM personal computer. It has a capacity of 8-bit and 16-bit. Despite being not fast enough for most of today's computer applications, the ISA bus is still used.

ii. Peripheral Component Interconnect (PCI) was originally built to fulfill video needs as a result of using the graphical user interface. The PCI bus has a capacity of 32-bit and 64-bit, which is 20 times faster than the ISA bus. The PCI is used to connect the CPU, memory, and expansion board.

iii. Accelerated Graphics Port (AGP) is the most recent bus with twice the speed of the PCI bus. The AGP is used specially for accelerating graphic performances. It is often used for graphics and 3-D animation, and also for replacing the PCI bus in video data transfers.

2.2 INPUT AND OUTPUT

2.2.1 Introduction

In this topic, you will learn about various input and output methods. The keyboard is the most popular input device used. Besides the keyboard, there are many other latest input and output devices that are available in the market. They will also be discussed here.

Input is like a bridge between data and processing, which is to enable data to be carried from the users to the computer. Input may be required to answer questions, and it needs processing. Output is like a link between computer processing and people. Output to be produced is determined by the analysis of users' needs. Input and output devices are also referred to as peripherals. We are going to see how input and output devices work, as shown in Figure 2.9 below.

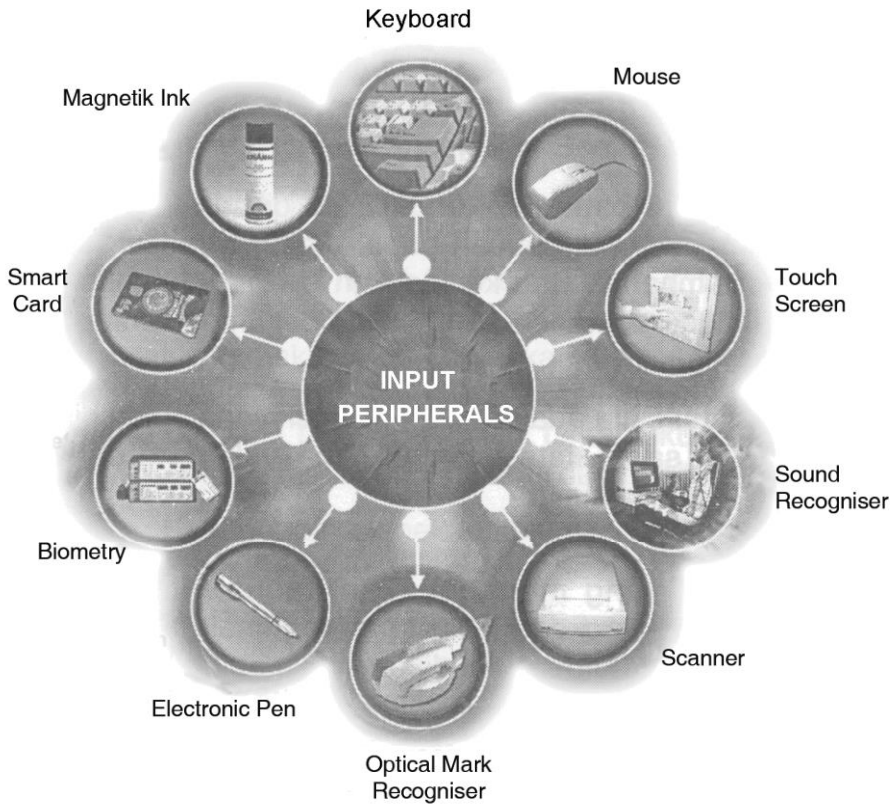


Figure 2.9: Input devices

2.2.2 Input Methods

Various types of source documents require various input methods and devices. Direct data entry consists of data that can be read by the machine (computer) and transferred as input directly into the computer. Indirect data entry requires various media and processing of the data source before the actual computer processing. In general, there are three approaches to data entry, as shown in Table 2.11.

Table 2.11: Types of Data Entry into Database

Types of Input Data	Definition
Off-line data entry	After data has been entered, it will be changed into a suitable form, and will be verified. Data will be corrected if there are errors. All data that has been confirmed as correct will be processed. This method is less often used now, and this is also referred to as the batch method.
On-line data entry	Similar to off-line data entry; the difference is that the data entered is directly examined and corrected. After all the data has been entered, processing will then be done on them.
Interactive data entry	Similar to on-line data entry, except that, data is examined every time it is entered and processing on the data is done immediately. There is no need to wait for all the data to be entered.

Input devices take data and programmes that can be read or understood by humans, and convert them into a form that can be processed by the computer. This new form consists of electronic signals of 0 and 1, which can be read by machines as explained in the section on data representation.

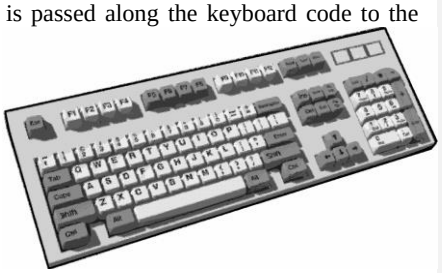
There are two types of input devices:

1) Entry via the keyboard: Data is transferred as input into the computer via a keyboard that resembles the keyboard of a typewriter, but contains a few additional keys. For this type of entry, users read the original document, called the source document. The contents of this document will be entered via the keyboard.
2) Direct entry: Data is already in a machine-readable form, and it is simultaneously entered into the computer, without using the keyboard.

An example of an input device that uses both types of entries, entry via the keyboard and direct entry, is the Point-Of-Sale (POS) terminal. It is a kind of electronic cash register that is used widely in business.

THE KEYBOARD

It is made up of a circuit board and related electronic components that generate a unique electronic code when each key is pressed. The code is passed along the keyboard code to the computer system unit where it is translated in a reusable form for processing. It is devices that are often used for data entry, and the terminal for displaying what has been entered. There are various forms, styles, sizes, arrangements, touch or feel, and the number of keys. The keyboard contains a control and buffer keys. Whatever is typed will be stored inside a buffer (which can store more than one symbol), and will later be scanned by the computer processor after receiving interrupt signals from the control key.



THE KEYBOARD FUNCTIONS OF THE SELECTED KEYS
The keyboard is split into district areas.

1. **THE TYPEWRITER AREA**

This area of the keyboard is similar to the standard typewriter. Most of the keys can repeat a character by simple holding down each key as needed.

The upper keys are those characters that required combinations of both shift and the specific key to produce the specific character and the lower keys are those that can just be pressed without the shift key to produce the specific character. Examples

Caret (^)	Tilde (~)	Examination point (!)	Number sign (#)	
Percentage sign (%)		Ampersand (&)	Braces ({})	Asterisk (*)
Under score (_)		Parenthesis ([ ])	Front slash (/)	Back slash (\)

2. **THE NUMERIC KEYPAD**

It will produce numbers only when the Num Lock light is on. The lower keys can only function when the Num Lock light is off. The Enter and Del keys are merely duplication keys found in the main typewriter part of the keyboard.

3. **THE CURSOR CONTROL KEYS**

These keys will always produce cursor control movement depending on the particular duplication being used.

4. **THE FUNCTION KEYS**

The function keys mark F1 to F12 may differ in uses in different application program. For instance, pressing F3 from MS-DOS prompt will repeat the last command entered. The other keys Print Screen, Scroll Lock and Pause or Brake may also use in different ways.

KEY COMBINATION

These are certain combinations of keys that can be used to performed special functions often normally character keys used in combination with either Alt (alternative) or Ctrl (Control) keys.

The Alt Keys

The Alt keys can be used to access the complete character set to your selected language, characters that are not available directory from individual keys. For example using the America keyboard and MS-DOS, holding down the Alt key and pressing the numbers from the numeric keypad will provide some special characters below:

- Alt 155 = ¢
- Alt 156 = £
- Alt 157 = ¥
- Alt 8358 = ₦
- Alt 0128 = €

The keys

- To move the cursor 8 space(s) = Tab key
- To cancel a procedure = Escape
- To correct a typing mistake at the left side of the cursor = Backspace
- To correct a typing mistake at the right side of the cursor = Del/Delete
- To create space between words = Space bar

- To type the upper case while the light is on = Shift
- To type the capital letters continuously = Caps Lock Light (on)
- To execute a command = Enter
- To move from one line to another = Enter
- To print or copy what is on the screen = Print screen
- To move twenty lines up = Page up
- To move twenty lines down = Page down
- To move the cursor to the beginning of the line = Home
- To move the cursor to the end of the line = End
- To switch between projector's views for an option = Windows key+P

Terminal

The terminal is an input (or output) device that connects you to a mainframe or other types of computers, called a host computer, or server. There are three types of terminals - dumb terminals, intelligent terminals, and network terminals.

Their functions are defined and shown in Table 2.13.

Types of Terminal	Definition
Dumb Terminals	They are used for input and/or to receive data, but cannot process data by themselves. They are used only to access information from the computer and are often found at airline companies, post offices, and other offices.
Intelligent Terminals	They have a processing unit, memory, and secondary storage like a magnetic disk. Basically, an intelligent terminal is a microcomputer with the communication software and modem or other communication connectivity. This connects the terminals to a bigger computer, or the Internet. The most popular type is NetPC.
Network Terminals	Also known as network computers, they are cheaper alternatives to intelligent terminals. Most network terminals do not have a hard disk drive, and must rely on the host computer or the server for their system and application software.

POINTING DEVICE

(i) Mouse: it is an input device for moving (rolling) to direct a pointer/cursor/icon on the screen. It's a small pointing device designed to fit comfortable under your hand. The mouse is normally used in GUI (windows) environments for opening and closing an icon, expanding and shrinking a window and choses command, but the keyboard is faster than the mouse in terms of operating. When the ball at the bottom of the mouse rolls, the vertical and horizontal coordinates (position) of the mouse cursor changes. By checking the coordinate, a computer can determine the cursor's position and displays the cursor on the screen. It is considered to be the most user-friendly device. The mouse is divided into two main types in terms of operation, the left mouse button and the right mouse button.

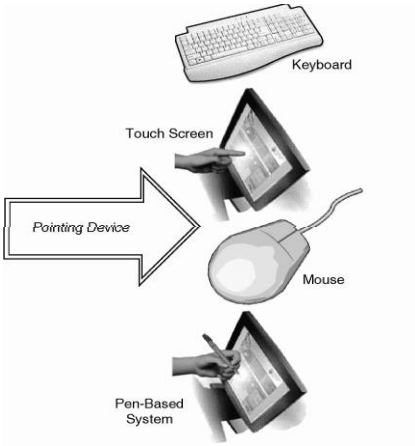


Figure 2.11: Pointing devices

Function of the Right Mouse Button

One alternative approach is to use the right mouse button on the desktop; the desktop is unlike a window, does not have a menu bar. Using the right mouse button on the desktop produces a Context Menu. The right mouse button provides a variety of menu options at a different location in windows.

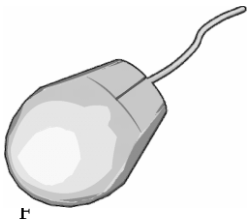


Figure 2.12: Mouse

Function of the Left Mouse Button

- ✓ For selecting a specific menu
- ✓ Enabling you to select an option from a menu or window
- ✓ Double clicking opens a folder, file, or application (icon)
- ✓ Moving an icon, window or an item from one location to another

Figure 2.13: Wireless Mouse

There are various models of the mouse as illustrated in Table 2.14. below:

Types of Mouse	**Definition**
Ball Mouse	It uses balls to move, and also known as a mechanical mouse
Optical Mouse	It uses light and requires a mouse pad that can reflect the light. This mouse contains a section called photo-detector. It functions like the ball mouse, and is normally used on a high-powered computer like the server.
Touch Sensitive Mouse	This has a pad that is sensitive to finger touch, for moving the icon/cursor.
Wireless Mouse	It communicates with the computer by emitting infrared signals or low-intensity radio signals.

(ii) **Touch Screen**: It is a monitor screen that is overlaid with a plastic layer. At the back of the layer, there is an invisible, criss-crossing infrared light. This arrangement enables a user to choose either an action or instruction on the screen by using fingers. The touch screen is easier to use, especially when someone requires information immediately. It is usually used at automatic teller machines (ATM), information kiosks, restaurants, and petrol stations.

(iii) Light Pen: It is a pen, or is a device like a pen, that is light sensitive and is connected via a cable to the computer. The location (or choice) is identified by pressing its button against a specific position on the screen. It is normally used for drawing images and selection of the menu.

(iv) Digital Board: It is a device that can be used to plot or to copy a certain drawing or picture. The item to be copied is placed on a digital tablet. Then, a specific stylus connected to the computer is used to plot the material. As the stylus moves from one location to another, the computer records the locations from the digital tablet. When the material has been plotted, the image can be displayed on screen, printed on paper, or stored in a computer system for future use. Digitisers are normally used by designers, architects, and engineers.

(v) Pen-Based System: This uses a pen to enter handwritings or marks into the computer containing software to identify and digitise handwritings and also hand drawings.

Scanning Device

Scanning devices translate images, texts, drawings, pictures, and similar items into a digital form for direct entry to the computer. Images that have been processed can be displayed on the screen, saved on the storage device, and sent to other computers. There are two technologies - **optical recognizer** and **magnetic recognizer**, as illustrated in Figure 2.14

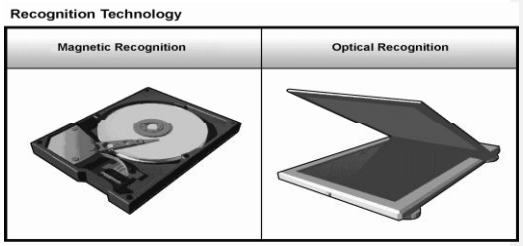
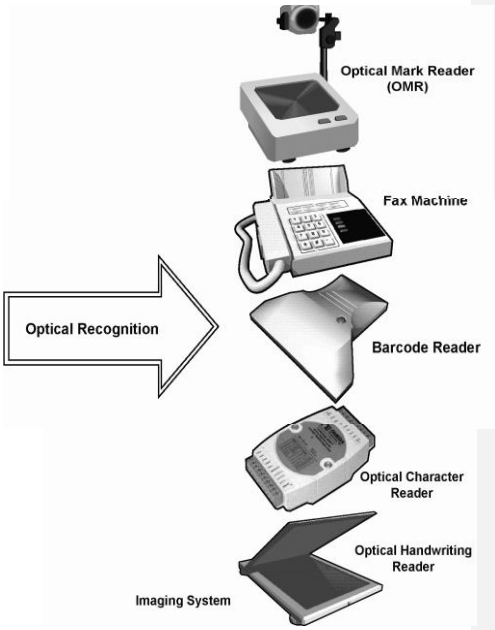


Figure 2.14: Recognition technology

1. **Optical Recognition:**

This technology-device, also known as the optical reader, uses light to scan images. An example of an optical recognition device is illustrated in Figure 2.15 below.



(a) Barcode Reader: For reading input in bar-form - marks that resemble vertically-shaped zebra stripes.

(b) Optical Mark Reader (OMR): Uses a device that reads pencil marks for objective questions of examinations. Such as, the technology used by WAEC for marking the objective questions of Section A.

(c) Optical Character Reader (OCR): For reading specific pre-printed characters (in specific font), for example the OCR-A font (the one perfected by ANSI).

(d) Optical Handwriting Reader: For reading handwriting, but its capability is still limited.

(e) Fax Machine: For scanning images and sending them electronically via telephone line to a destination (fax machine).

There are two types:

● Dedicated Fax Machine: Or simply known as the fax machine.

● Fax Modem: It is installed in a computer to enable electronic documents to be sent directly to its receiver (either fax machine or computer). It is faster but it cannot scan non-electronic documents.

Figure 2.16: Fax machine

(f) Imaging System: It is an image scanning device that converts all texts, drawings, pictures, and others into digital form. Imaging system technology has created imaging industries by combining the use of scanners, cameras, etc.

2. **Magnetic Recognition:**

This uses magnetic methods to scan magnetic writing. The technological device used is the Magnetic Ink Character Reader (MICR). The MICR reads the number (in strange form) that is printed using magnetic ink. The MICR is often used in financial sectors to facilitate cheques sorting, as done by the Bank of Ghana, and other banks.

Voice Recognition Device:

It converts human voice/speech into digital codes. Human voice is transferred through a microphone that is connected to a computer. When this input device is combined with sound cards and appropriate software, a system of sound recognition is formed. This system enables a user to operate a microcomputer to develop documents by giving orders using the human voice. It has the potential to be used as a security method.

Smart and Optical Card:

The smart card refers to the multi-purpose card, or MyKad, distributed by the National Registration Department, as well as credit and debit cards distributed by various commercial banks such as StandChart and Ghana Commercial Bank. The use of the optical card is still restricted despite its ability to store information of up to 2,000 pages as compared to smart cards (able to store information of up to 30 pages, depending on the capability of the microprocessor), and magnetic cards (able to store information of up to half a page).

Other Input Devices

There are other input devices that can receive input in the form of audio, video, electronic pictures, senses and human biology. Most of these tools enable the use of multimedia, and there are some used for security objectives, such as the concept of biometry. We can see various other input devices in Figure 2.17 below.

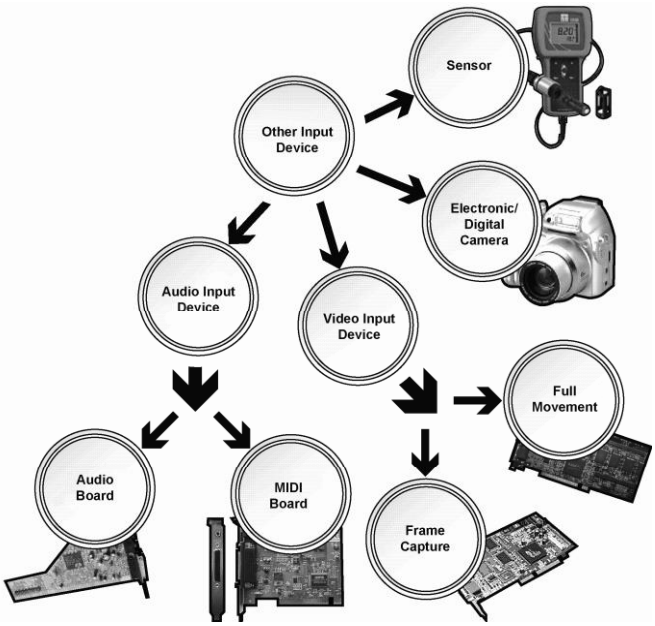


Figure 2.17: Other input devices

(i) **Audio Input Device**: It is used to record or play analogue music into digital form. It can generally receive various forms of sound as input, whereas, a sound recognition device can receive only one type of audio input - i.e. sound input. There are two methods of digitising the audio:

(a) Audio Board: The board that changes analogue sound to digital sound and saves it for future use.

(b) MIDI Board: Musical Instrument Digital Interface (MIDI) board emits digital music, which normally uses the MIDI keyboard.

(ii) **Video Input Device:** Enables video pictures to be converted into digital form through a specific card. There are two types of video card:

(a) Frame Capture: Can digitise one frame at a time only.

(b) Full Movement: Can convert at the rate of 30 frames per second.

The main problem here is storage, not the input. Nowadays, digital video cameras are available in the market, where the output is in digital form. Digital video recorders will facilitate multimedia presentation or a form of new broadcasting on the Internet, i.e. web television.

(iii) **Electronic/Digital Camera**: It is similar to the traditional camera except that images are recorded digitally in the camera, or inside the camera memory, and not on film. You can snap a picture, look at it, and place it on the website within a few minutes.

(iv) **Sensor:** These are input devices that collect data in specific forms directly from the environment and send them to the computer. Sensors can be used to track data such as speed, weight, temperature, image, shape, light, air pollution, etc. Have you ever weighed yourself using an audio-enabled weighing scale at a supermarket? That is one example of a sensing device used to measure your weight, and also your height.

2.2.3 Input Control

There must be input control to safeguard the integrity of data and to prevent them from becoming Garbage in Garbage out (GIGO). Input control is a collection of manual and computer control methods that have been designed to ensure all input data are actually precise when entered into the computer. Now, you have understood why information system needs to be developed.

Commented [o5]: Outline 2 characterizes of hardcopy and softcopy each
Which of them would you recommend for a multinational company and why

2.2.4 Output Methods

Output can take various forms - such as screen output, printed-paper, cathode ray, signals, voice, microfilm, etc. In general, there are two types of output:

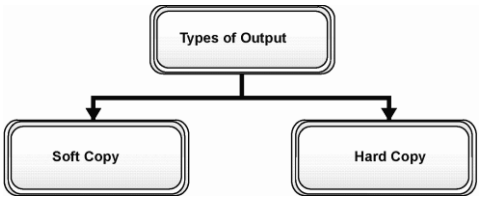


Figure 2.19: Types of output

(i) Soft Copy

This refers to a non-permanent copy of data shown on the display screen, in audio or voice form, or in a stored electronic form in secondary storage. This type is most suitable for data or information that changes constantly, and is always updated electronically. Its information can only be read or transfer when the computer is on or active. It can also be distributed quickly in real-time across the globe. Also, multiples copies can be produced at a time. It also provides security features that prevent unauthorised users for access. This copy is suitable for multinational organisation or business due to its real-time for disseminating data and information. Some of the drawbacks are access can only be done with power, the content can be destroyed by Very Important Resources Under Siege (VIRUS), and hackers can still the model.

(ii) Hard Copy
This refers to a permanent copy of data in a printed form. Anything that comes in printed form (whether text, graphic, or microfilm) are considered hard copies. This type of output is most suitable for data or information that does not change quickly. This copy is accessible without power at any time and any location. There some advantages and disadvantages that can be discussed.

Besides soft and hard copies, decisions on output largely depend on a number of factors, such as:
(a) The frequency of output required - whether always, regular, upon request or unpredictable.
(b) Cost involved.
(c) How much output to be produced.
(d) How fast the information is required.
(e) The situation of users.

2.2.5 Output Devices

Data entered needs to be processed by the computer so that they are in a machine-readable form until the output device converts them into a form that can be read by humans. Output devices used together with microcomputers are the monitor, printer, plotter, and voice output device. We can see examples of output devices as illustrated in Figure 2.20.

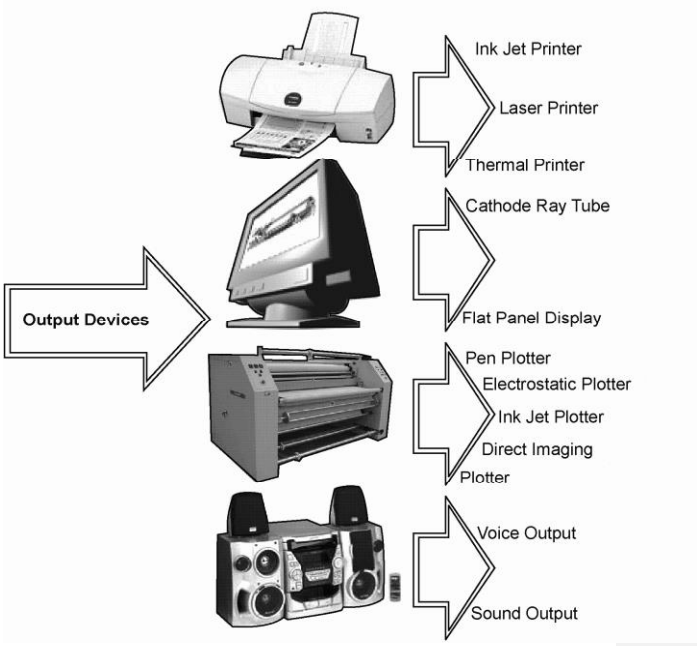


Figure 2.20: Output devices

Printer
The image displayed on the monitor is often referred to as the soft copy. Information that is produced on paper, whether from printer or plotter, is called a hard copy. Three popular types of printers always used together with microcomputers are ink jet, laser, and thermal printers.

(i) Ink Jet Printer: Sends out droplets of ink at high speed onto the surface of paper. This process does not only produce images of high-quality letters, but also allows printing to happen in various colours. An ink jet printer is a printer that is reliable, quiet and cheap.

(ii) Laser Printer: Utilises the emission of a laser beam onto the drum, and is transferred onto paper by using the toner (as in the photocopy machine). There is a laser printer that is classified as a PostScript printer. PostScript is a printer language, often called Page Description Language (PDL), which has been accepted as a standard for laser printer.

Figure 2.21: Printer

(iii) Thermal Printer: Uses heat element to produce images on heat-sensitive paper. Originally, this printer is used in scientific laboratories to record data. This printer is used specifically for producing output that is almost the same as photographs.

The measure of printer speed depends on the type of printer or the method of printing, i.e. whether on the basis of characters, lines or pages. The speed is characters per second, lines per minute or pages per minute. Most printer models are desktop printers which are difficult move around (depending on size). The use of laptop computers leads to portable printers. Among the criteria in choosing printers are image quality, speed, level of noise, and cost of operation such as ink cost, spare parts, etc.

Monitor

Monitors are purposely used for output. Some monitors also accept input with a special tool, such as pen and figure. There are two types of monitors - cathode ray tube and flat panel display.

(i) Cathode Ray Tube (CRT): It is a vacuum tube that is used as a display screen for the computer or video display terminal. The technology is the same as the television. The CRT is an output device commonly used together with the computer. The smallest unit on the screen is called a pixel (picture element).

Figure 2.22: Cathode ray tube monitor

(ii) Flat Panel Display: It is lighter, thinner, and uses less power as compared to the CRT. This panel is made of two pieces of glass/plastic that have activated materials among them. The clarity of a monitor is determined by the resolution which is measured in pixels. A pixel is a dot or "picture element" that forms an image on the monitor.

Figure 2.23: Flat Panel Display

For a certain monitor size, higher resolution (more pixels) leads to clearer image. For a certain degree of clarity, a bigger monitor requires a higher resolution.

Standard

A number of standards have been developed for determining the degree of computer resolutions. Four standards that are commonly used now are SVGA, XGA, SXGA and UXGA. The definitions of these standards are as shown in Table below

Table 2.16: Computer Resolution Standards

Standard	**Definition**
SVGA	A short form for Super Video Graphics Array, it has a minimum resolution of 800 x 600 pixels
XGA	A short form for Extended Graphics Array, it has a resolution of up to 1,024 x 768 pixels. XGA is a popular standard now.
SXGA	Representing Super Extended Graphics Array, this standard has a resolution of 1,280 x 1,024 pixels.
UXGA	A short form for Ultra Extended Graphics Array, it is the latest and highest standard. Its use is continuously growing, especially for engineering designs at high level and for graphical fine arts.

Plotter
Plotters are used specifically for producing bar charts, maps, architectural drawings, and three-dimensional illustrations. A plotter can produce multicoloured documents of high quality, and also documents bigger than that produced by other printers. There are four types of plotters: pen, ink jet, electrostatic, and direct imaging.

Voice Output Device
There are two types of technology, i.e. voice-output technology (speech coding & synthesising), and sound-output technology (FM synthesising & virtual acoustic).

1. Voice Output: A voice output device converts digital data into sound, such as speech. There are two voice-output technologies, as illustrated in Figure 2.23

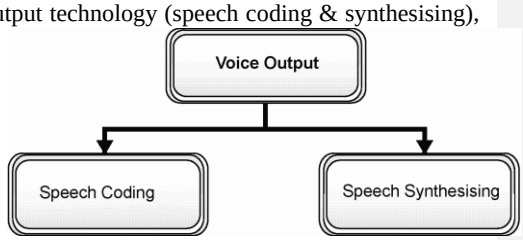


Figure 2.23: Voice output

(a) Speech Coding: It uses the human voice that speaks certain words and is stored in digital form. After accessing the voice, it will re-produce the words of the spoken voice
(b) Speech Synthesising: It uses a total of 40 phonemes (suitable for English) as the basis for speech in electronic form to create any word.

This method does not use the human voice. Computers convert texts into voice,

2. Sound Output: A sound output device produces digital sounds, such as beeping and music. It requires software and a voice card or digital audio board (like the Sound Blaster). There are two technologies - i.e. FM synthesis and virtual acoustic, as illustrated in Figure 2.24. In both cases, sound output will be sent to a mixer before being sent to a speaker.

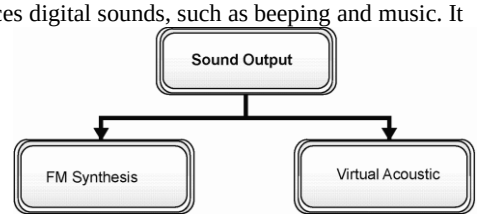


Figure 2.24: Sound output

(a) FM Synthesis: It is a synthesiser that mimics various musical instruments by using stored sound - unlike the original musical instrument.
(b) Virtual Acoustic: A method that does not use stored sound, but stores software that models certain real musical instruments. It is important for multimedia applications.